

# Aetherminds — Neuromorphic Motor Intent Detection

## NeuroNex'26 Hackathon Project

A spike-inspired, low-power Brain-Computer Interface (BCI) system that detects motor intent from EEG signals using a Leaky Integrate-and-Fire (LIF) Spiking Neural Network — designed to assist stroke patients in controlling assistive devices.

## Problem Statement

Stroke patients often lose voluntary motor control. Traditional EEG-based BCI systems are computationally expensive and impractical for real-time, low-power deployment. This project applies **neuromorphic computing principles** — mimicking how biological neurons communicate via sparse spikes — to build an efficient, low-power motor intent classifier directly from EEG signals.

## How It Works

```
Raw EEG Data (MNE PhysioNet EEGBCI)
    ↓
Preprocessing & Normalization
    ↓
LIF Spike Encoding (spikingjelly)
    ↓
Feature Extraction
(spike rate · count · variance · sparsity)
    ↓
Random Forest Classifier (scikit-learn)
    ↓
Motor Intent → Assistive Device Trigger
(LEFT HAND / RIGHT HAND)
```

## Neuromorphic Math

### LIF Membrane Update:

$$U[t+1] = \beta \cdot U[t] + W \cdot I[t] - V_{th} \cdot O[t]$$

where  $\beta = e^{(-\Delta t / \tau)}$  is the leak factor,  $V_{th}$  is the firing threshold.

### Spike Sparsity:

$\text{Sparsity} = 1 - (\text{\#spikes} / \text{\#total slots})$

High sparsity → most synapses stay quiescent → lower dynamic power vs dense neural networks.

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## Project Structure

```
Aetherminds-EEG-signals2026/
├── SNN.py           # LIF spike encoding + feature extraction + model
training
├── Demo.py          # Runs inference on test samples (terminal demo)
├── Visualize.py     # Generates EEG spike raster visualizations
├── app.py           # Flask web API – /predict/<sample_id>
├── templates/
│   └── index.html   # Web dashboard frontend
├── requirements.txt  # Python dependencies
└── LICENSE          # MIT License
```

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## Setup & Installation

### Prerequisites

- Python 3.10+
- pip

### 1. Clone the Repository

```
bash

git clone https://github.com/sudish-666/Aetherminds-EEG-signals2026.git
cd Aetherminds-EEG-signals2026
```

### 2. Create Virtual Environment

```
bash

python -m venv venv
source venv/bin/activate      # Linux/Mac
# venv\Scripts\activate      # Windows
```

### 3. Install Dependencies

```
bash

pip install -r requirements.txt
```

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### ▶ Running the Project

**Important:** Run in this order — SNN.py must run first to generate the trained model and data files.

#### Step 1 — Train the SNN Model

```
bash

python SNN.py
```

Downloads the MNE EEGBCI motor imagery dataset, encodes EEG epochs into LIF spike trains, extracts features, and trains the classifier. Generates `x_data.npy`, `y_labels.npy`, `model.pkl`, and `scaler.pkl` locally.

#### Step 2 — Run Terminal Demo

```
bash

python Demo.py
```

Runs inference on 5 test samples and prints motor intent predictions:

```
=====
MOTOR INTENT DETECTED: RIGHT HAND
Inference Latency: 821.059ms
Spike Sparsity: 62.2% (low power mode)
ASSISTIVE DEVICE TRIGGERED → RIGHT HAND MOVEMENT
=====
```

#### Step 3 — Generate Visualizations

```
bash

python Visualize.py
```

Produces EEG spike raster plots comparing Left Fist vs Right Fist motor intent signals across 20 EEG channels.

## Step 4 — Launch Web API

```
bash

python app.py
```

Open `http://localhost:5000` in your browser. Use `/predict/<sample_id>` endpoint to get real-time motor intent predictions.

### Demo Results

| Sample | Detected Intent | Latency (ms) | Spike Sparsity |
|--------|-----------------|--------------|----------------|
| 1      | RIGHT HAND      | 821.059      | 62.2%          |
| 2      | LEFT HAND       | 1148.640     | 56.7%          |
| 3      | LEFT HAND       | 1094.544     | 55.5%          |
| 4      | RIGHT HAND      | 1068.699     | 60.2%          |
| 5      | RIGHT HAND      | 1088.613     | 50.6%          |

**Average Spike Sparsity ~57%** — meaning ~57% of neurons remain inactive per inference, significantly reducing dynamic power consumption compared to conventional dense neural networks.

### Tech Stack

| Component       | Technology                    |
|-----------------|-------------------------------|
| EEG Dataset     | MNE-Python · PhysioNet EEGBCI |
| Spike Encoding  | spikingjelly (LIF neurons)    |
| Deep Learning   | PyTorch                       |
| Classifier      | scikit-learn (Random Forest)  |
| Web API         | Flask                         |
| Visualization   | matplotlib                    |
| Data Processing | NumPy · SciPy · pandas        |

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## Key Features

- **Neuromorphic spike encoding** — maps raw EEG amplitude changes to binary spike events, mimicking biological neural communication
  - **High sparsity inference** (~50-62%) — enables potential deployment on neuromorphic hardware such as Intel Loihi or IBM TrueNorth
  - **Real-time latency** under 1.2 seconds — feasible for assistive device control loops
  - **No GPU required** — LIF encoding + Random Forest runs entirely on CPU
  - **Web API included** — Flask backend serves live predictions via REST endpoint
- 

## Real-World Impact

Designed for stroke rehabilitation: the system detects imagined hand movements from scalp EEG and can trigger prosthetic limbs, robotic exoskeletons, or smart wheelchairs — giving stroke patients greater autonomy without invasive brain surgery.

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## License

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## Author

**Sudish Agarwal** GitHub: [@sudish-666](#) Project: [Aetherminds-EEG-signals2026](#)